

Xu Han

CS GRADUATE STUDENT, HUAZHONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Visiting Student, King Abdullah University of Science and Technology, Saudi Arabia

✉ xhanxu@hust.edu.cn | 🏠 xhanxu.github.io | 📄 GitHub | 🎓 Google Scholar | *Make it count.*

Education

King Abdullah University of Science and Technology (KAUST)

Kingdom of Saudi Arabia

VISITING STUDENT, CEMSE

Jun. 2025 - Dec. 2025

- Supervised by **Prof. Peter Wonka**.
- Conducted research on text-to-intrinsic generation, proposing a unified diffusion model to efficiently synthesize intrinsic maps from text.

Huazhong University of Science and Technology (HUST)

Wuhan, China

MASTER OF SCIENCE (M.Sc.), COMPUTER SCIENCE, SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2023 - Jun. 2026 (expected)

- Supervised by **Prof. Xianzhi Li**.
- **GPA: 3.91 (3/166)**, National Scholarship, Tencent Scholarship, BYD Scholarship, First Prize Scholarship, Innovation Scholarship.

Shandong University (SDU)

Qingdao, China

BACHELOR OF ENGINEERING (B.Eng.), ARTIFICIAL INTELLIGENCE, SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2019 - Jun. 2023

- Supervised by **Prof. Mengbai Xiao**, Institute of Intelligent Computing.
- **GPA: 3.87 (88.7)**, Honours Degree (1/52), National Scholarship (Top 0.2% nationwide), Outstanding Thesis (Top 6 grads in CS, 2%).

Research Interest

My research lies at the intersection of **vision & graphics**, **representation learning**, and **generative modeling**, with a focus on **structured and efficient world modeling**. I am broadly interested in how **structured inductive biases** can make representation models and generative systems more coherent, controllable, and interpretable.

1. **World Modeling: Structured Representation Learning and Generative Models**
2. **Multimodal Learning: Physically Grounded Intrinsic Understanding and Generation**
3. **Efficient Architectures: Structured Matrix, Sparsity, Tensor Factorization**

Ultimately, my goal is to develop **structured, principled generative systems** that unify perception and synthesis, enabling models to understand not only *how the world looks* but also *how it works*.

Publication

[1] LumiX: Structured and Coherent Text-to-Intrinsic Generation

Submitted to CVPR 2026

XU HAN, BIAO ZHANG, XIANGJUN TANG, XIANZHI LI, PETER WONKA

[Paper](#) [GitHub](#)

- We present LumiX, a structured diffusion framework for coherent text-to-intrinsic generation. Conditioned on text prompts, LumiX jointly generates a comprehensive set of intrinsic maps (e.g., albedo, irradiance, normal, depth, and final color), providing a structured and physically consistent description of an underlying scene.

[2] MoST: Efficient Monarch Sparse Tuning for 3D Representation Learning

CVPR 2025

XU HAN, YUAN TANG, JINFENG XU, XIANZHI LI

[Paper](#) [GitHub](#)

- We introduce Monarch Sparse Tuning (MoST), the first reparameterization-based parameter-efficient fine-tuning (PEFT) method tailored for 3D point cloud representation learning.

[3] Mamba3D: Enhancing Local Features for 3D Point Cloud Analysis via State Space Model

ACM MM 2024

XU HAN*, YUAN TANG*, ZHAOXUAN WANG, XIANZHI LI (*EQUAL CONTRIBUTION)

[Paper](#) [GitHub](#)

- We present Mamba3D, a state space model tailored for point cloud learning. Mamba3D surpasses existing methods in multiple tasks, achieving multiple SoTA, with only linear complexity.

[4] More Text, Less Point: Towards 3D Data-Efficient Point-Language Understanding

AAAI 2025

YUAN TANG*, XU HAN*, XIANZHI LI†, QIAO YU, JINFENG XU, YIXUE HAO, LONG HU, MIN CHEN (*EQUAL CONTRIBUTION,

[Paper](#) [GitHub](#)

† CORRESPONDING AUTHOR)

- We introduce a new task, 3D Data-Efficient Point-Language Understanding. Our proposed GreenPLM uses text data to compensate for the lack of 3D data, achieving superior 3D understanding with only 12% or even without 3D data.

[5] PointDreamer: Zero-Shot 3D Textured Mesh Reconstruction From Colored Point Cloud

TVCG 2025

QIAO YU, XIANZHI LI, YUAN TANG, XU HAN, JINFENG XU, LONG HU, YIXUE HAO, MIN CHEN

[Paper](#) [GitHub](#)

- We propose PointDreamer, a framework that adapts 2D diffusion models to 3D point clouds via a novel project-inpaint-unproject pipeline, achieving superior texture quality over prior text- or image-driven methods.

[6] Fancy123: One Image to High-Quality 3D Mesh Generation via Plug-and-Play Deformation

CVPR 2025

QIAO YU, XIANZHI LI, YUAN TANG, **XU HAN**, JINFENG XU, LONG HU, YIXUE HAO, MIN CHEN

[Paper](#) [GitHub](#)

- We propose a SOTA framework for single-image-to-3D-mesh, leveraging 2D deformation, 3D deformation, and unprojection to resolve multiview inconsistency, low fidelity, and blurry coloration.

[7] SASep: Saliency-Aware Structured Separation of Geometry and Feature for Open Set Learning on Point Clouds

CVPR 2025

JINFENG XU, XIANZHI LI, YUAN TANG, **XU HAN**, QIAO YU, YIXUE HAO, LONG HU, MIN CHEN

[GitHub](#)

- We introduce Saliency-Aware Structured Separation (SASep), an open-set recognition method on 3D point cloud.

[8] MiniGPT-3D: Efficiently Aligning 3D Point Clouds with Large Language Models using 2D Priors

ACM MM 2024

YUAN TANG, **XU HAN**, XIANZHI LI[†], QIAO YU, YIXUE HAO, LONG HU, MIN CHEN ([†] CORRESPONDING AUTHOR)

[Paper](#) [GitHub](#)

- We present MiniGPT-3D, an efficient and powerful 3D-LLM that aligns 3D points with LLMs using 2D priors. It has only 47.8 M learnable parameters and is trained in just 26.8h on a single RTX 3090.

[9] patchDPCC: A Patchwise Deep Compression Framework for Dynamic Point Clouds

AAAI 2024

ZIRUI PAN, MENGBAI XIAO[†], **XU HAN**, DONGXIAO YU, GUANGHUI ZHANG, YAO LIU ([†] CORRESPONDING AUTHOR)

[Paper](#)

- We propose patchDPCC to compress each frame of the point cloud video by divides frames into patch groups, and incorporate a feature transfer module to refine the feature quality.

Experience

King Abdullah University of Science and Technology (KAUST)

Saudi Arabia

STUDENT RESEARCHER, SUPERVISED BY **PROF. PETER WONKA**.

Jun. 2025 - Now

- Research focused on **structured content generation**, with an emphasis on **text-to-intrinsic generation**, in collaboration with Prof. Peter Wonka, Dr. Biao Zhang, and Dr. Xiangjun Tang.
- Led the development of **LumiX**, a unified diffusion model that produces pixel-aligned geometry, material, and lighting representations, enabling photorealistic multi-map synthesis, PBR modeling, and world modeling.

Huazhong University of Science and Technology

Wuhan, China

RESEARCH ASSISTANT, SUPERVISED BY **PROF. XIANZHI LI**.

Sep. 2023 - Now

- Led research on **efficient 3D representation learning**, focusing on scalable and structured architectures, resulting in two major outcomes: **Mamba3D** (ACM Multimedia) and **MoST** (CVPR).
- As a core contributor, conducted research on **multimodal large language models** aligning 3D representations with language and vision, resulting in **MiniGPT-3D** (ACM Multimedia) and **GreenPLM** (AAAI).

Institute of Intelligent Computing, Shandong University

Qingdao, China

RESEARCH ASSISTANT, SUPERVISED BY **PROF. MENGBAI XIAO**.

Oct. 2020 - Jun. 2023

- Developed a dynamic point cloud upsampling model to reduce bandwidth consumption in point cloud video streaming by aligning adjacent frames in feature space. This work received the **Outstanding Graduation Thesis Award** and was further extended to point cloud video compression, resulting in a publication at **AAAI 2024**.

Honors & Awards

SCHOLARSHIPS

10/2025	National Scholarship , Highest honor for postgraduates, top 0.2% nationwide	Wuhan, China
10/2025	Research & Innovation Scholarship , Two-year continuous, HUST	Wuhan, China
03/2025	Tencent Scholarship , Two-year continuous, HUST	Wuhan, China
01/2025	BYD Scholarship , The only one in Dept. of CS, HUST	Wuhan, China
10/2024	Xiaomi Scholarship Nomination , HUST	Wuhan, China
10/2024	Research & Innovation Scholarship , HUST	Wuhan, China
04/2024	Tencent Scholarship , HUST	Wuhan, China
11/2023	First Prize Scholarship , HUST	Wuhan, China
10/2022	National Scholarship , Highest honor for undergraduates, top 0.2% nationwide	Qingdao, China
2021,2022	Huawei Scholarship , Two-year continuous	Qingdao, China
10/2022	Second Prize Scholarship , Top 10% in Department of Computer Science	Qingdao, China
10/2022	Research & Innovation Scholarship , Shandong University	Qingdao, China

AWARDS

10/2025	Merit Student , HUST	Wuhan, China
01/2025	Best Paper Award, HUST School of CS Annual Conference , Top 10 in School of CS	Wuhan, China
07/2023	Outstanding Graduation Thesis Award , Top 6 graduates in Department of Computer Science	Qingdao, China
06/2023	Honours Bachelor Degree , 1/52	Qingdao, China
06/2023	Outstanding Graduates Award , Shandong University	Qingdao, China
2021,2022	Huawei-MOE (Ministry of Education) Future Star Award , Two-year continuous	Qingdao, China
11/2021	First Prize in China Undergraduate Mathematical Contest in Modeling , Top 0.6% in 45K teams	Qingdao, China